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Georgia Institute of Technology
Aerospace Engineering

B.S. Aerospace Engineering
Expected May 2027

M.S. Aerospace Engineering
Expected May 2028



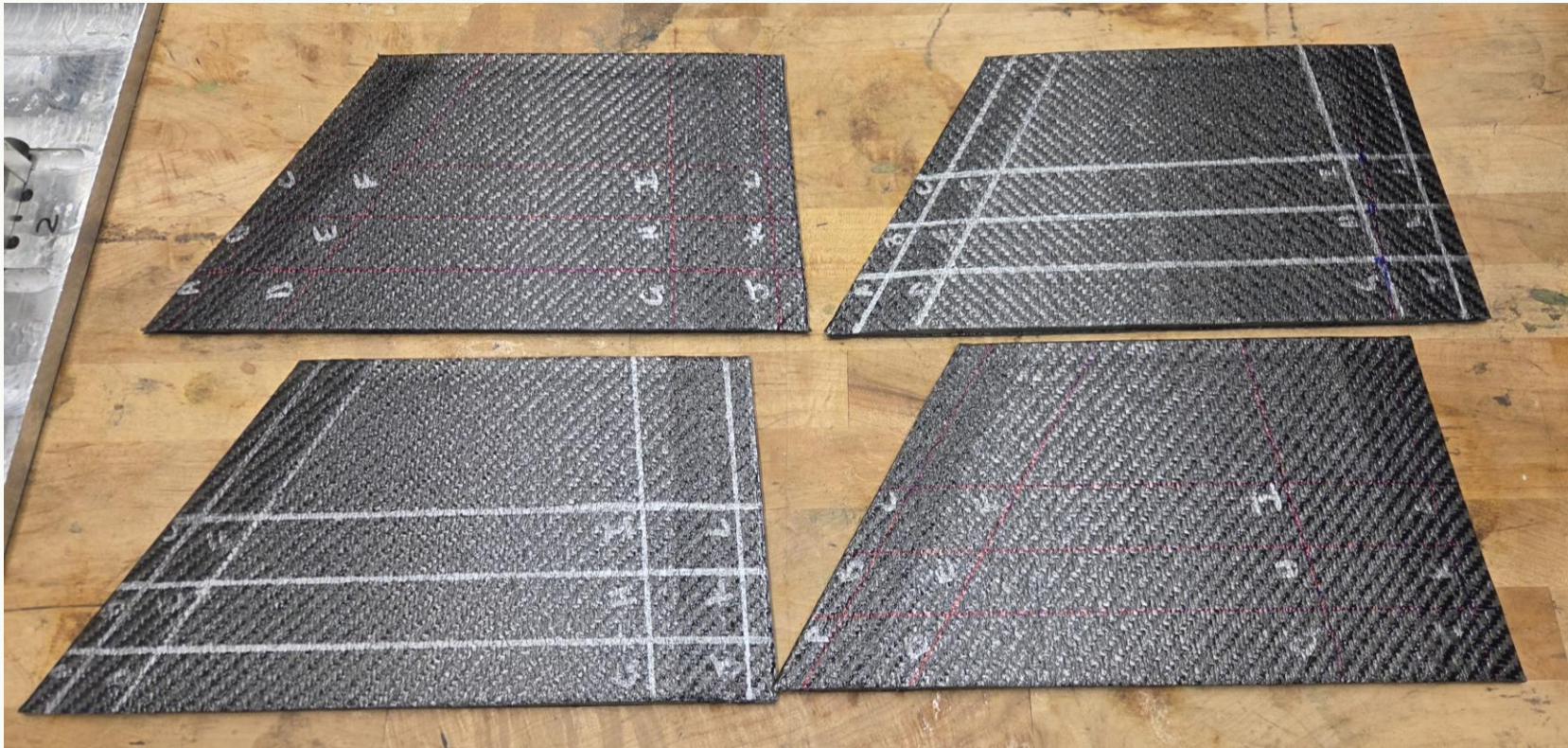
Pursuit
AEROSPACE



Georgia Tech Experimental Rocketry

Fin Design & Manufacturing

GTXR / FIN DESIGN & MANUFACTURING



Finished fins ready for attachment

Requirements Drove High Level Fin Geometry

I developed a structured trade study comparing aerodynamic performance, flutter margin, and manufacturing constraints.

Determining Geometry

Hexagonal

Selected for reproducible geometry and higher predicted flutter margin.

Short taper

Selected: easier machining and higher flutter margin.

Long taper

Lower aerodynamic penalty, but difficult to machine.

Double wedge

Aerodynamically preferred, but manufacturing criteria were not met.

Full taper

Rejected with the double-wedge concept.

Full G10 Selected for Manufacturing Confidence

The short-taper trade favored a repeatable process over the lower mass of a framed foam core.

Material Selection

SHORT-TAPER CORE	ADVANTAGE	LIMITATION	DECISION
Full G10	Simple, repeatable manufacturing	Highest mass of these options	Selected
G10 frame + Rohacell	Lowest mass of these options	Foam-forming process not mature	Rejected
Aluminum frame + Rohacell	Easier machining; thermal resistance	CTE mismatch; prior interface delamination	Rejected

1.72 lb

full G10

0.48 lb

G10 frame + Rohacell

+1.24 lb

accepted mass difference

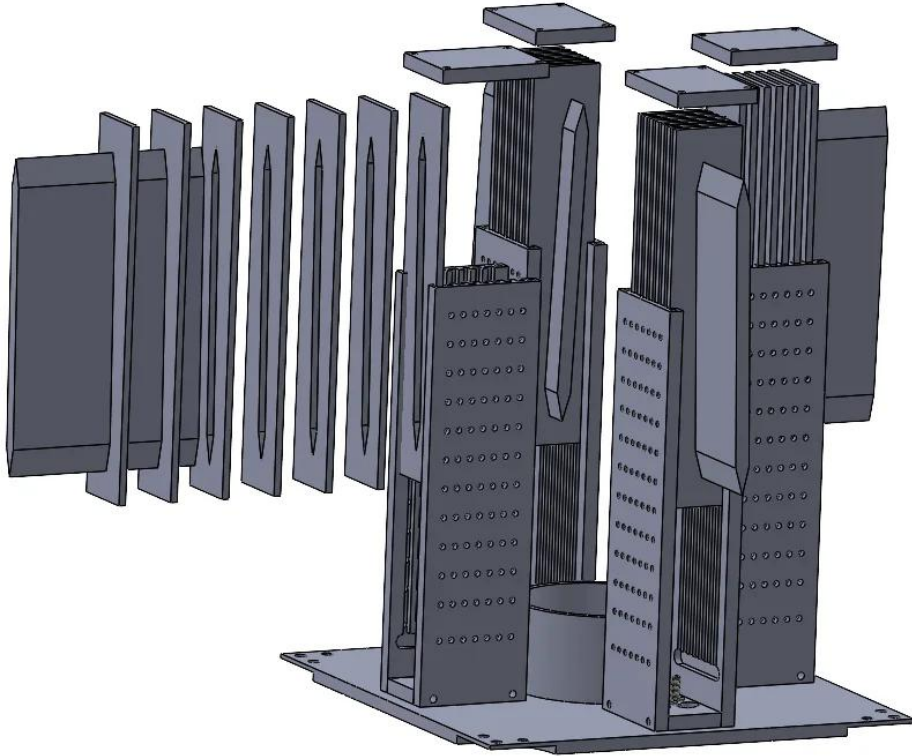
Georgia Tech Experimental Rocketry Fin Alignment Tool

GTXR / FIN DESIGN & MANUFACTURING

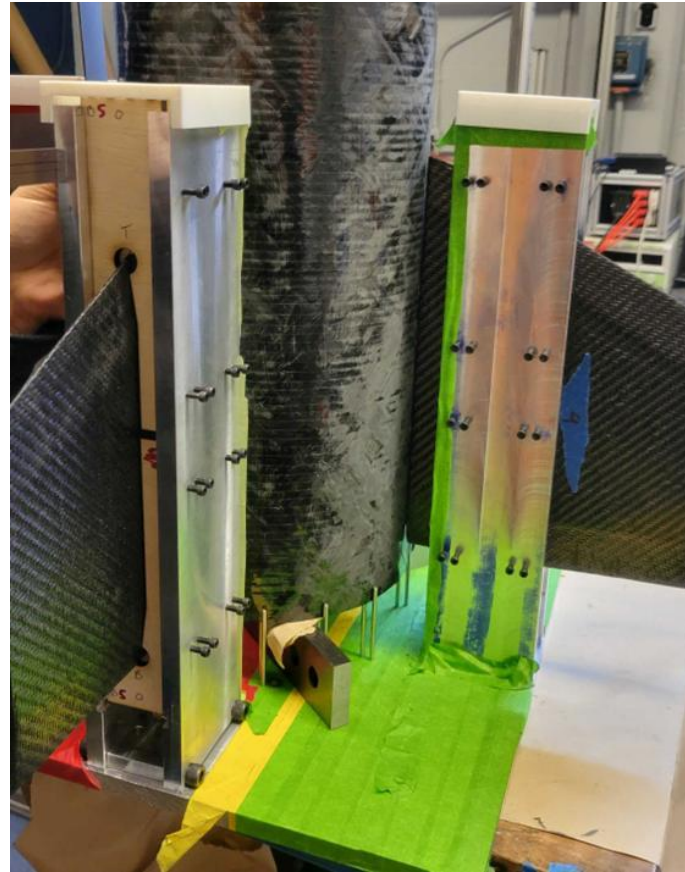
Notice the slight
1.6° angle



Modular Tooling Establishes a 1.6° Fin Cant



SolidWorks model



Fixture constrains fins during attachment

Design requirement

Achieve $1.6^\circ \pm 0.1^\circ$ cant while accommodating different rocket diameters and fin designs.

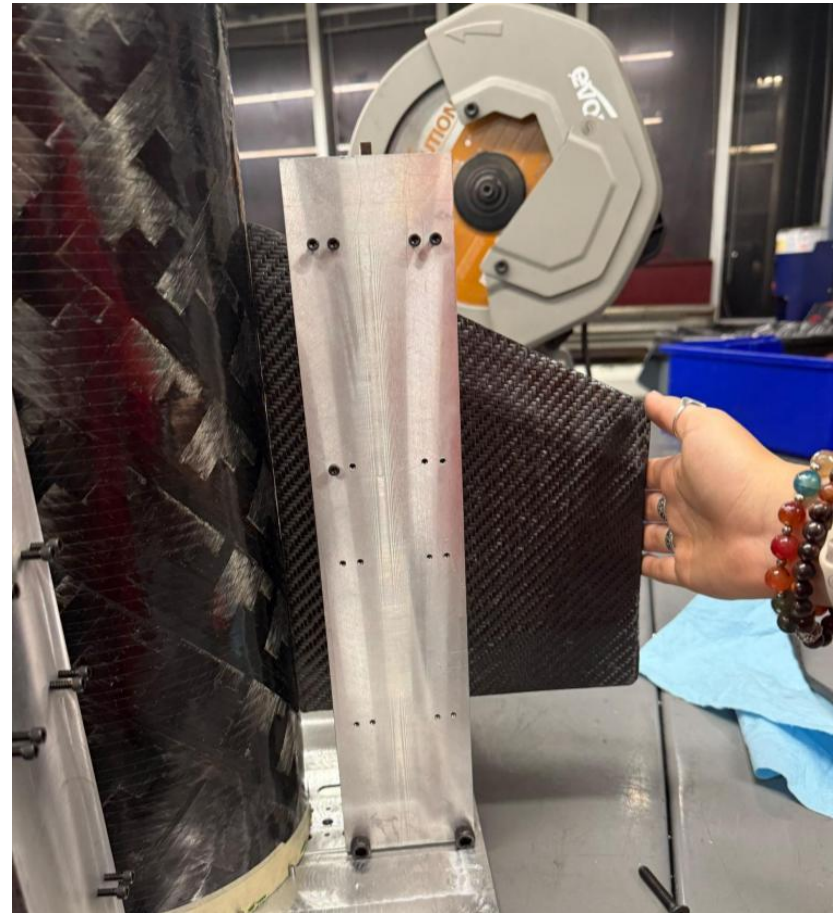
My contribution

Led fixture design and development of the operating procedure. Aided in CAM programming and machining all hardware.

Manufacturing and Assembling the Alignment Fixture



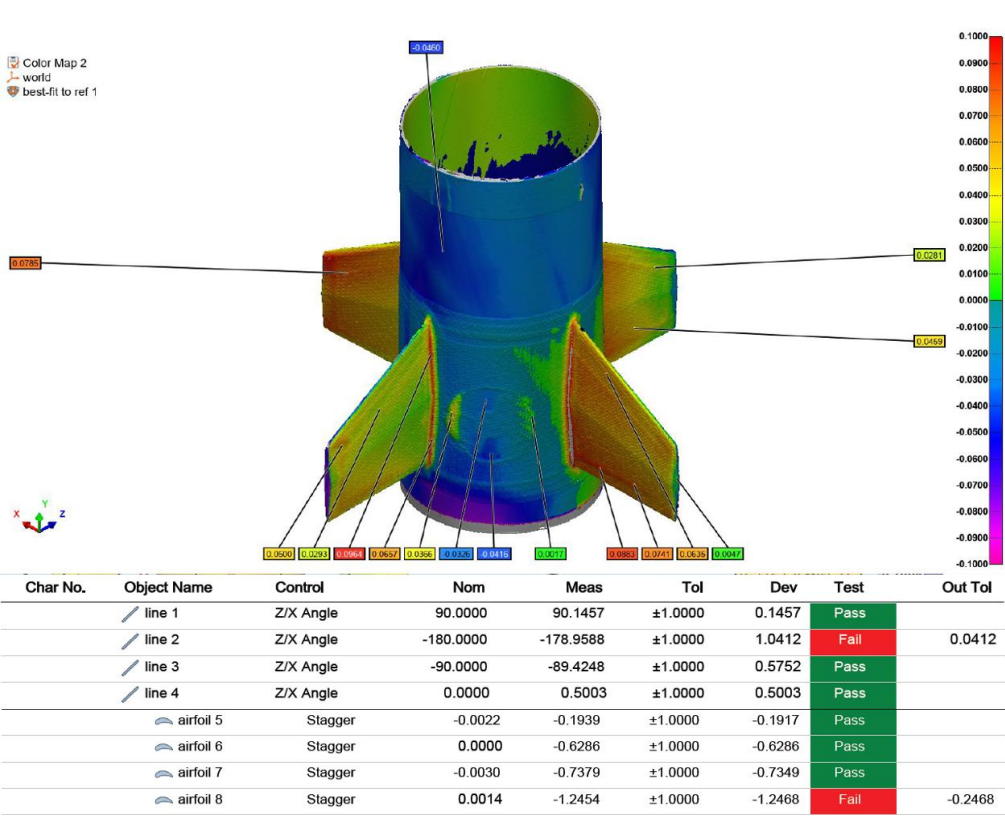
Machined components



Fixture constrains fins during attachment.

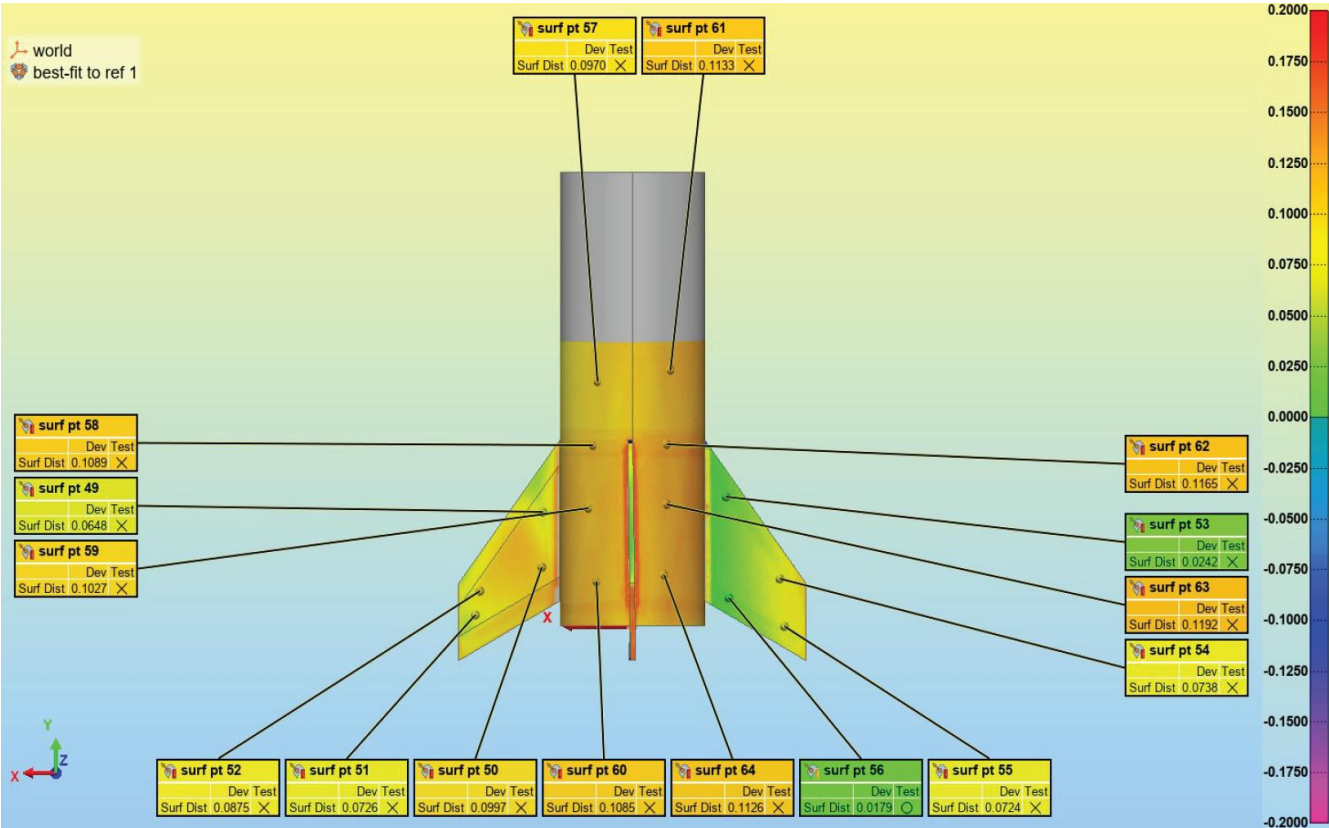
Independent Metrology Verified the Tool - HEXAGON

Before tool



Multiple fins failed alignment criteria.

After using tool



The team reported no dimensional failures.

<0.1° deviation from the 1.6° target on all four fins

GTXR Thermoforming Foam Research

GTXR / MANUFACTURING RESEARCH

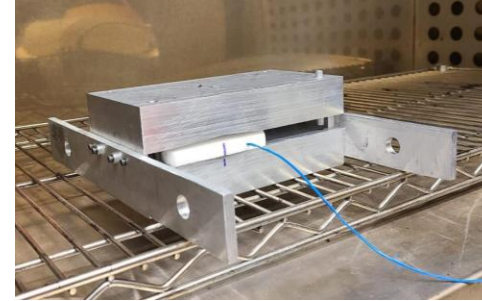


Fin shaped thermoformed PMI foam

Heat, Form, and Cool a Double-Wedge Foam Core

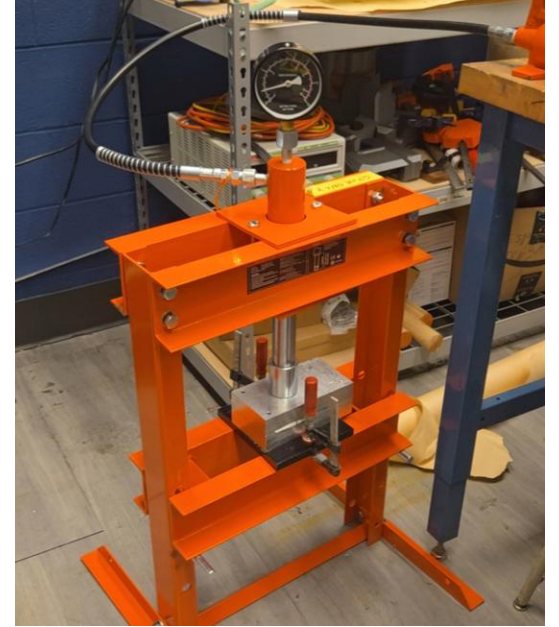


Aluminum tooling defines the candidate fin-core geometry.



195°C

Target forming temperature



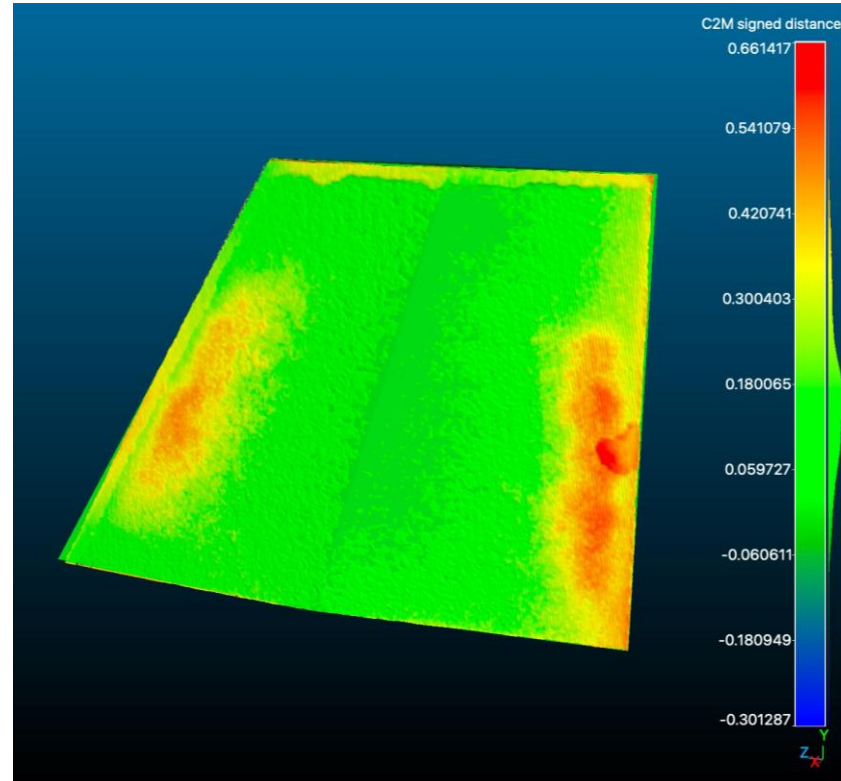
Co-developed the process and machined tooling. Heated the PMI blank, closed the mold, and cooled under load.

Co-author and presenter, AIAA 2026-112793

Measured Geometry Exposed Process Limitations



3D Scanner for Metrology Data Collection



3D scan compared with nominal geometry.

0.28 mm

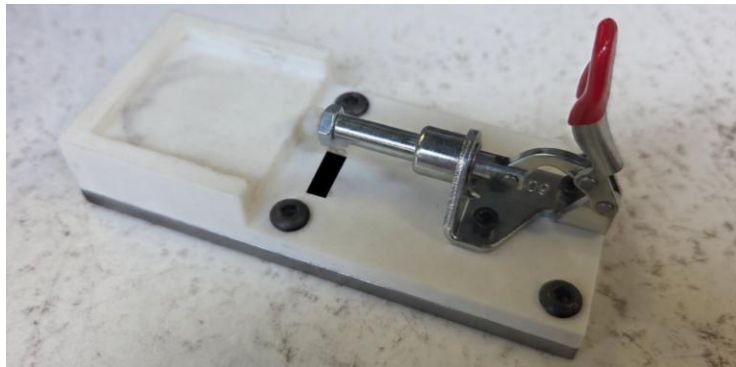
average global geometric error across five cores

Metrology identified local deformation and error areas for manufacturing defect analysis.

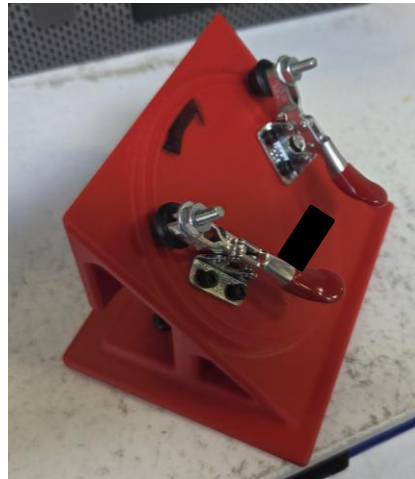
Manufacturing feasibility research. The PMI cores were not yet viable for flight.

The process required improved constraint and repeatability before flight use.

Quick-Change Fixtures Reduced Production Handling



Marking fixtures. Identifiers removed.



15

quick-change fixtures developed & manufactured

83%

Average reduction in load/unload time

95%

reduction in inspection time using Go/No-Go gauges I designed (not pictured)

Emergency Passthroughs Enabled Stage Ignition



Finished NPT passthroughs

Developed electrical passthrough hardware for the second-stage ignition system.

14 hours

Conception → design → fabrication
turnaround time

The hardware flew
successfully and supported
second-stage ignition

2× MEOP

Achieved during hydrostatic testing

This design informed
subsequent passthrough
hardware

A Better Mock Rail – V2



V2 rail designed by the GTXR intern, shown during all-up practice.

Observed V1 limitations

The rail flexed, was not straight, and did not hold the rocket steady.

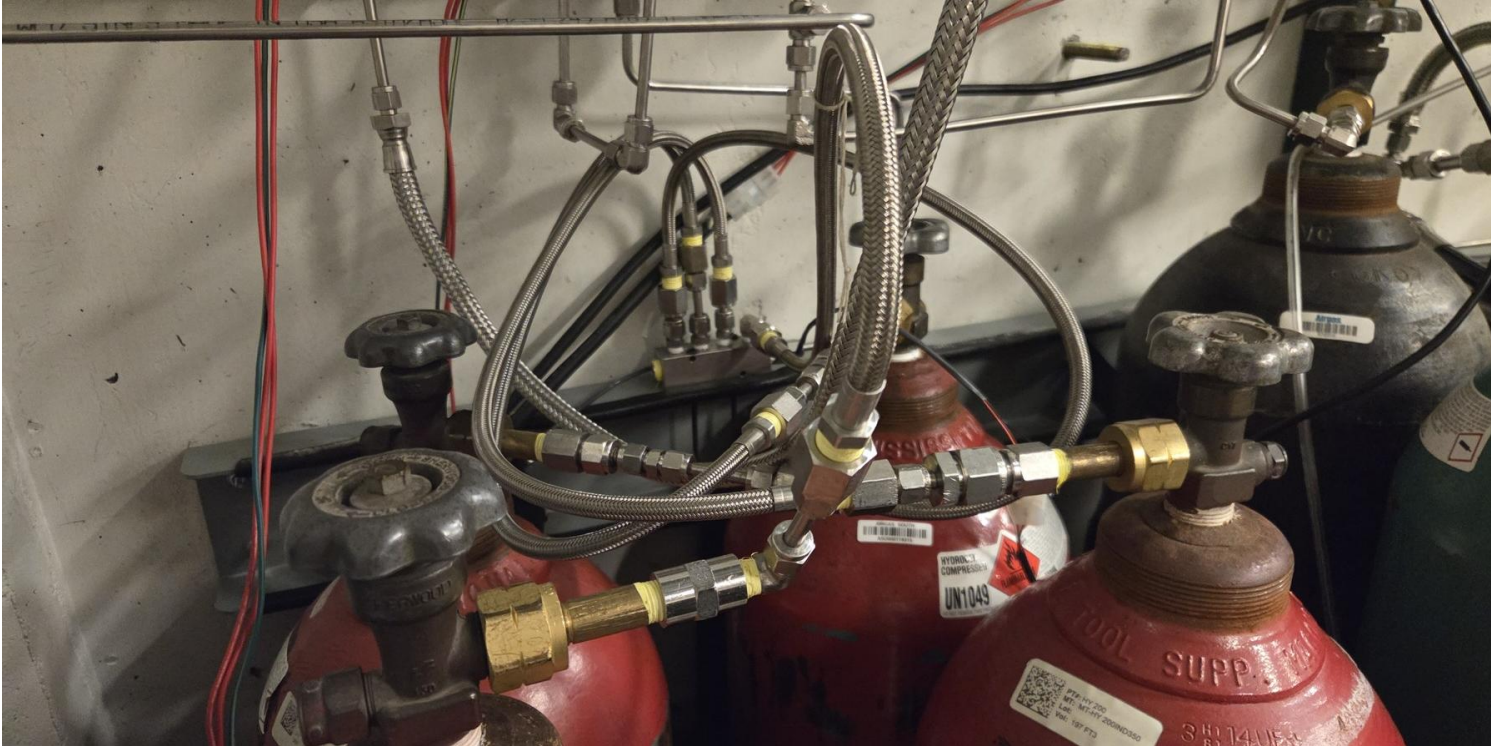
My contribution

Recommended an I-beam and larger base with the structures lead. Aided design and load-path analysis.

- Bending and Euler buckling checks guided the design.

Implemented in V2, with stay cables for alignment and raising.

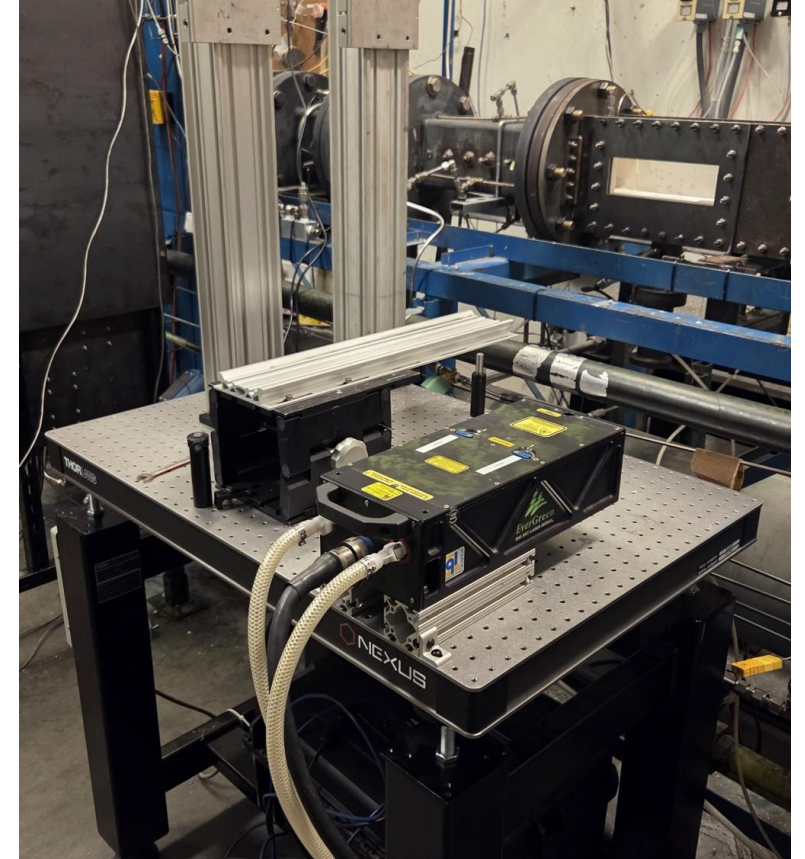
Built fluid routing and supported laser PIV



Routed tubing to combine four hydrogen bottles.

Completed piping work and helped assemble experimental hardware for combustion research and flow measurements.

Nathaniel Murcin



Helped construct the laser PIV setup.

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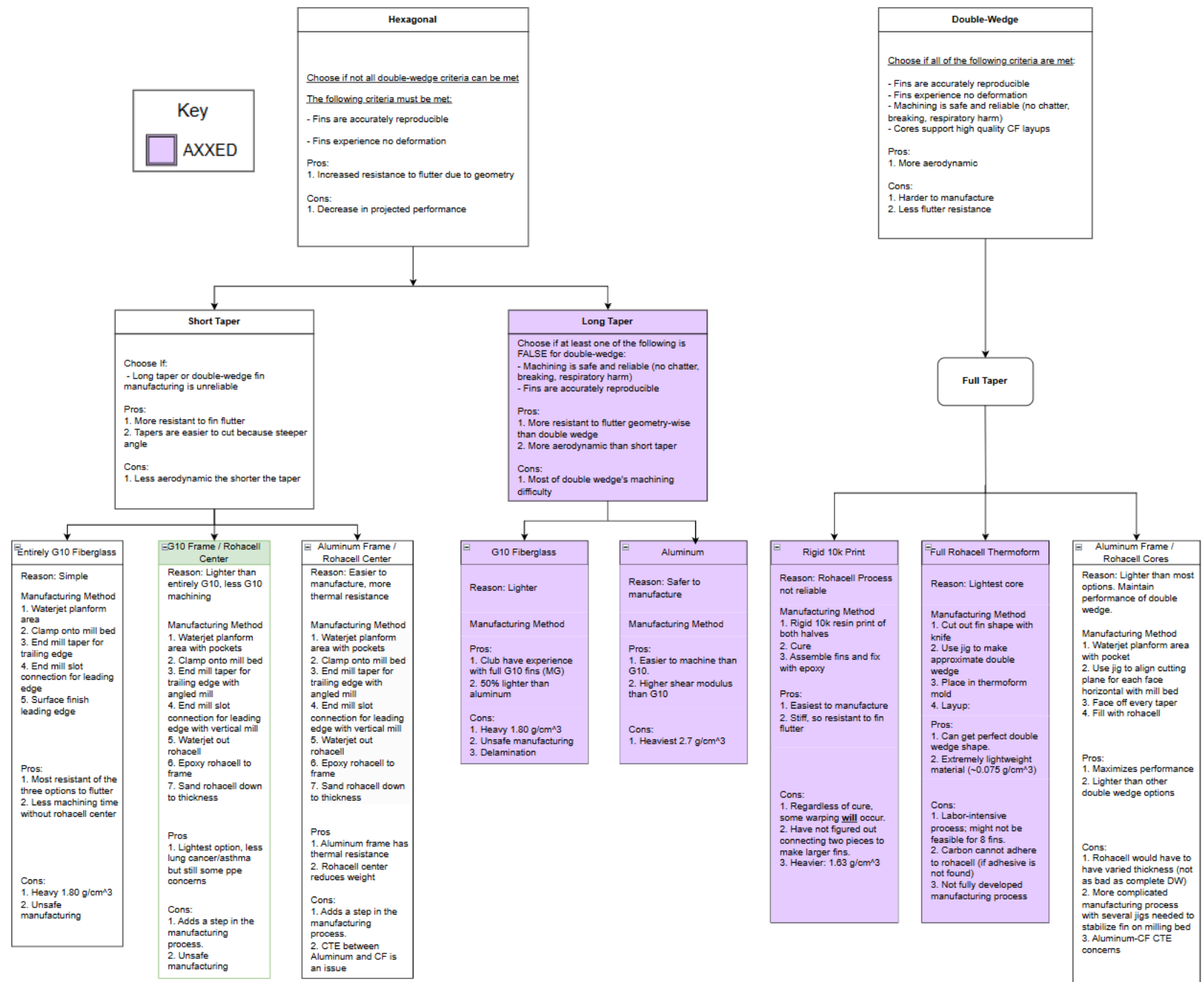
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[nmurcin.github.io](https://github.com/nmurcin)



Space Shuttle Atlantis, Kennedy Space Center

Appendix

Full fin decision matrix



V1 Mock Rail

